

Assignment 1

Deadline: Tuesday, 5th of May 2026

Upload your solutions at: <https://tinyurl.com/AML-2026-ASSIGNMENT1>

1. **(0.3 points)** What is the maximum value of the natural even number n , $n = 2m$, such that there exists a hypothesis class $\mathcal{H}$ with n elements that shatters a set C of $m = \frac{n}{2}$ points? Give an example of such an $\mathcal{H}$ and C . Justify your answer.

2. **(1.2 points)** Consider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$, where:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x \geq a]}(x) = \mathbf{1}_{[a, +\infty)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x < a]}(x) = \mathbf{1}_{(-\infty, a)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_{a,b} : \mathbb{R} \rightarrow \{0, 1\} \mid h_{a,b}(x) = \mathbf{1}_{[a \leq x \leq b]}(x) = \mathbf{1}_{[a,b]}(x), a, b \in \mathbb{R}\}.$$

Consider the realizability assumption.

a) Compute $\text{VCdim}(\mathcal{H})$.

b) Give an algorithm A and determine an upper bound on the sample complexity $m_{\mathcal{H}}(\epsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

3. **(1.2 points)** Usually, there is a correlation between the number of parameters of a hypothesis class $\mathcal{H}$ and its VC-dimension. Consider the following hypothesis classes:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[a, a+1]}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_b : \mathbb{R} \rightarrow \{0, 1\} \mid h_b(x) = \mathbf{1}_{[b, b+1] \cup [b+2, b+4]}(x), b \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_c : \mathbb{R} \rightarrow \{0, 1\} \mid h_c(x) = \mathbf{1}_{[c, c+1] \cup [c+2, c+4] \cup [c+6, c+9]}(x), c \in \mathbb{R}\}.$$

Consider the realizability assumption. Compute $\text{VCdim}(\mathcal{H}_1)$, $\text{VCdim}(\mathcal{H}_2)$, $\text{VCdim}(\mathcal{H}_3)$.

Ex-officio: 0.3 points